

The Measurement and Structure of the National Economy

ECON202 MACROECONOMICS-1

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Learning Objectives

1. Differentiate among the three approaches to national income accounting

By the end of this Module, you should be able to

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4. Explain the calculations of real GDP, price indexes, and inflation

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2. Explain how GDP is measured
3. Discuss the measurement of aggregate saving and its relation to wealth
4. Explain the calculations of real GDP, price indexes, and inflation
5. Define real and nominal interest rates

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Roadmap of Lecture

National Income Accounting

Gross Domestic Product

- * Juice Business Example: Calculation

Saving and Wealth

Real GDP, Price Indexes, and Inflation

- * Inflation
- * Application

Interest Rates

Problems

National Income Accounting (1 of 3)

- **National income accounts:** an accounting framework used in measuring current economic activity
- **Three alternative approaches give the same measurements**
 - **Product approach:** the amount of output produced
 - **Income approach:** the incomes generated by production
 - **Expenditure approach:** the amount of spending by purchasers

Each approach gives a different perspective on the economy. However, the fundamental principle underlying national income accounting is that, except for problems such as incomplete or misreported data, all three approaches give identical measurements of the amount of current economic activity.

National Income Accounting (2 of 3)

- **Juice business example** shows that all three approaches are equal
- **Important concept in product approach:**
Value added = value of output minus value of inputs purchased from other producers



National Income Accounting (3 of 3)

Why are the three approaches equivalent?

- They must be, by definition
- Any output produced (product approach) is purchased by someone (expenditure approach) and results in income to someone (income approach)
- **The fundamental identity of national income accounting:**

$$\text{total production} = \text{total income} = \text{total expenditure} \quad (2.1)$$

1. Identity: A statement that is always true by definition. It's true because of how we define and measure the variables involved. **a dozen = 12 items** - it's true because that's how we define a dozen.
2. Equation: A statement that may or may not be true depending on economic conditions. It represents a relationship that needs to be tested or verified.

$$C = a + bY \quad (1)$$

3. This is an equation because it represents a hypothesis about behavior. It suggests that consumption depends on income in a specific way, but this relationship needs to be tested with real data. The values of 'a' and 'b' aren't given by definition - they're discovered through research.

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The Product Approach

- **GDP (gross domestic product)** is the market value of final goods and services newly produced within a nation during a fixed period of time

1. what happens in the financial markets and to public opinion on the day a new GDP report comes out. 1. Either only measure final goods and services or 2. Only measure the value added at each stage of production as a good moves from firm to firm to final purchaser.
2. when measuring the size of a local economy, common sense and economic sense are likely to conflict. Common sense says, "Measure the size of the local economy by adding up the sales of all the local businesses." But that would include massive double counting. Economic sense says something different: "Measure the size of the local economy by summing up the value added by each local business." To do that, you need to know the cost of each company's outputs and inputs and then just sum all the values of the outputs while subtracting the sum of the inputs.

The Product Approach (2-3 of 14)

- **Market value:** allows adding together unlike items by valuing them at their market prices
 - **Problem:** misses nonmarket items such as homemaking, the value of environmental quality, and natural resource depletion
 - There is some adjustment to reflect the underground economy
 - Government services (that aren't sold in markets) are valued at their cost of production
- **Newly produced:** counts only things produced in the given period; excludes things produced earlier

1. Analytical Problems 1 and 3 both discuss difficulties in counting nonmarket items for GDP, including the important idea that GDP is not the same as welfare.
2. WHAT IS NOT IN GDP
 - household production
 - non-market transactions (barter)
 - natural resources depleted by production
 - leisure, inequality, health, social/political liberties
 - new indicators are developed to better measure well-being
 - gross national happiness (GNH) in Bhutan since 1972
 - <https://www.nytimes.com/2017/01/17/world/asia/bhutan-gross-national-happiness-indicator-.html>

The Product Approach

■ Final goods and services

- Don't count intermediate goods and services (those used up in the production of other goods and services in the same period that they themselves were produced)
- Final goods & services are those that are not intermediate
- Capital goods (goods used to produce other goods) are final goods since they aren't used up in the same period that they are produced

1. A reputable study shows that a particular new workplace safety regulation will reduce the growth of real GDP. Is this an argument against implementing the regulation? Explain.
2. Capital goods (machinery, equipment, buildings) are treated as final goods in GDP calculation, not intermediate goods.
3. Capital goods are produced in one period but used over multiple periods. Since they aren't "used up" in the same year they're made, we count their full value as final output in the production year.
4. A \$50,000 machine made in 2024 will produce goods for 10 years → Count full \$50,000 in 2024 GDP.
5. Prevents undercounting productive investment and accurately captures economy's capacity-building in the current period.

The Product Approach

- **Final goods and services**

- **Inventory investment** (the amount that inventories of unsold finished goods, goods in process, and raw materials have changed during the period) is also treated as a final good
- Adding up value added works well, since it automatically excludes intermediate goods

GNP vs GDP

- **GNP (gross national product)** = output produced by domestically owned factors of production
- **GDP** = output produced within a nation
- $GDP = GNP - NFP$ (2.2)
- **NFP** = net factor payments from abroad = payments to domestically owned factors located abroad minus payments to foreign factors located domestically

GNP vs GDP

- **Example:** Engineering revenues for a road built by a U.S. company in Saudi Arabia is part of U.S. GNP (built by a U.S. factor of production), not U.S. GDP, and is part of Saudi GDP (built in Saudi Arabia), not Saudi GNP
- Difference between GNP and GDP is small for the United States, about 0.2%, but higher for countries that have many citizens working abroad

The expenditure approach

- Measures total spending on final goods and services produced within a nation during a specified period of time
- Four main categories of spending: consumption (C), investment (I), government purchases of goods and services (G), and net exports (NX)

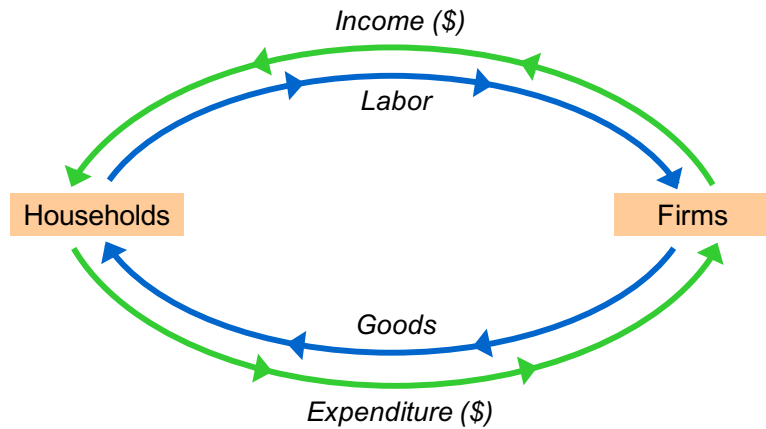
$$Y = C + I + G + NX \quad (2.3)$$

the income-expenditure identity

The expenditure approach

- **Consumption:** spending by domestic households on final goods and services (including those produced abroad)
- About $\frac{2}{3}$ of U.S. GDP
- **Three categories**
 - Consumer durables (examples: cars, TV sets, furniture, major appliances)
 - Nondurable goods (examples: food, clothing, fuel)
 - Services (examples: education, health care, financial services, transportation)

Income=expenditure



The expenditure approach

- **Investment:** spending for new capital goods (fixed investment) plus inventory investment

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- **Inventory investment:** increases in firms' inventory holdings

The expenditure approach

- **Government purchases of goods and services:** spending by the government on goods or services

of U.S. GDP, Most by state and local governments, not federal government
Not all government spending counts toward GDP's "G" component.
Included in G: Government purchases of goods/services (salaries, infrastructure, equipment) - represents current production.
Excluded from G: Transfer payments (Social Security, unemployment benefits, welfare) - just redistributes money without creating new output. Transfer payments don't reflect current economic production, so including them would overstate GDP.
If government spends \$100B total but \$30B is transfers → Only \$70B counts in GDP calculation.

1. Some government spending is for capital goods that add to the nation's capital stock, such as highways, airports, bridges, and water and sewer systems

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 - Some are payments that are **not** made in exchange for current goods and services
 - One type is transfers, including Social Security payments, welfare, and unemployment benefits
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- Some government spending $\Rightarrow$ capital goods

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The expenditure approach

- **Net exports:** exports minus imports
- **Exports:** goods produced in the country that are purchased by foreigners
- **Imports:** goods produced abroad that are purchased by residents in the country
- Imports are subtracted from GDP, as they represent goods produced abroad, and were included in consumption, investment, and government purchases

T2.1: Expenditure Approach to Measuring GDP

	Billions of dollars	Percent of GDP
Personal consumption expenditures (C)	15,750	68.5
Consumer durables	2025	8.8
Nondurable goods	3457	15.0
Services	10,269	44.7
Gross private domestic investment (I)	4108	17.9
Business fixed investment	3051	13.3
Nonresidential structures	579	2.5
Equipment	1273	5.5
Intellectual property products	1200	5.2
Residential investment	1083	4.7
Inventory investment	−26	−0.1
Government purchases of goods and services (G)	4052	17.6
Federal	1565	6.8
National defense	905	3.9
Nondefense	660	2.9
State and local	2487	10.8
Net exports (NX)	−916	−4.0
Exports	2480	10.8
Imports	3396	14.8
Total (equals GDP) (Y)	22,994	100.0

Note: Numbers may not add to totals shown owing to rounding.

Source: Bureau of Economic Analysis website, www.bea.gov, NIPA Table 1.1.5, January 27, 2022.

1. Businesses reduced their stockpiles by \$26B - they sold more than they produced.
2. Simple example: A car dealer starts with 100 cars, produces 80 new cars, sells 90 cars → ends with 90 cars. Inventory decreased by 10 cars.
3. Subtracts from GDP because those extra sales came from previous years' production, not current output.
4. Why it happens: Strong consumer demand exceeding current production, or businesses clearing excess stock.
5. Economic signal: Either very strong current demand or companies becoming cautious about future sales.

The Income Approach

- Adds up income generated by production (including profits and taxes paid to the government)
- **National income** = compensation of employees (including benefits) + proprietors' income + rental income of persons + corporate profits + net interest + taxes on production and imports + business current transfer payments + current surplus of government enterprises
- National income + statistical discrepancy = net national product
- Net national product + depreciation (the value of capital that wears out in the period) = gross national product (GNP)
- $\text{GNP} - \text{net factor payments (NFP)} = \text{GDP}$

The Income Approach to measuring GDP (14 of 14)

Private sector and government sector income

- **Private disposable income** = income of the private sector = private sector income earned at home (Y or GDP) and abroad (NFP) + payments from the government sector (transfers, TR , and interest on government debt, INT) – taxes paid to Govt. (T)

$$= Y + NFP + TR + INT - T \quad (2.4)$$

- **Government's net income** = taxes – transfers – interest payments = $T - TR - INT$ (2.5)

- **Private disposable income + government's net income** = $GDP + NFP = GNP$

Table 2.2: Income Approach to Measuring GDP

	Billions of dollars	Percent of GDP
Compensation of employees	12,581	54.7
Proprietor's income	1822	7.9
Rental income of persons	726	3.2
Corporate profits	2806	12.2
Net interest	686	3.0
Taxes on production and imports	1148	5.0
Business current transfer payments	164	0.7
Current surplus of government enterprises	– 13	– 0.1
Total (equals National Income)	19,920	86.6
<i>Plus</i> Statistical discrepancy	– 520	– 2.3
<i>Equals</i> Net National Product	19,401	84.4
<i>Plus</i> Consumption of fixed capital	3848	16.7
<i>Equals</i> Gross National Product (GNP)	23,249	101.1
<i>Less</i> Factor income received from rest of world	1145	5.0
<i>Plus</i> Payments of factor income to rest of world	893	3.9
<i>Equals</i> Gross Domestic Product (GDP)	22,996	100.0

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Source: Bureau of Economic Analysis website, www.bea.gov, NIPA Tables 1.7.5 and 1.12, March 30, 2022.

Juice business example

Orangelnc Transactions

Wages paid to Orangelnc employees	\$15,000
Taxes paid to government	5,000
Revenue received from sale of oranges	35,000
Oranges sold to public	10,000
Oranges sold to Juicelnc	25,000

Juicelnc Transactions

Wages paid to Juicelnc employees	\$10,000
Taxes paid to government	2,000
Oranges purchased from Orangelnc	25,000
Revenue received from sale of orange juice	40,000

Approach 1: Product (Value-Added) Approach

Concept: Sum the value added at each stage of production

Company	Total Output	Purchased Inputs	Value Added
OrangeInc	\$35,000		
JuiceInc	\$40,000		
Total GDP			

- What inputs did OrangeInc purchase from other businesses?
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- Calculate value added for each company
- What is total GDP?

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Key Point: We subtract intermediate goods (\$25,000 oranges) to prevent double counting!

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Wrong approach: $\$35,000 + \$40,000 = \$75,000$ (counts oranges twice)

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Approach 2: Income Approach

Method A - Before-tax incomes : Sum all income earned by factors of production (wages + profits)

Income Source	Orangelnc	Juicelnc	Total
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Profits			
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Wages			
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Total GDP			
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Approach 2: Income Approach

Method A - Before-tax incomes : Sum all income earned by factors of production (wages + profits)

Income Source	OrangeInc	JuiceInc	Total
Profits	\$20,000	\$5,000	
Wages			
Total GDP			

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Total GDP			\$50,000

Approach 2B: Income Approach (Alternative)

After-tax incomes + government revenue:

Income Source	Amount
OrangeInc after-tax profits	
JuiceInc after-tax profits	
Total wages (no worker taxes)	
Government tax revenue	
Total GDP	

Calculate:

- OrangeInc after-tax profits = \$20,000 -
- JuiceInc after-tax profits = \$5,000 -
- Government taxes =

Approach 2B: Income Approach (Alternative)

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Total GDP	

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Income Source	Amount
OrangeInc after-tax profits	\$15,000
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Total wages (no worker taxes)	\$25,000
Government tax revenue	\$7,000
Total GDP	

Calculate:

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- JuiceInc after-tax profits = $\$5,000 - \$2,000 = \$3,000$
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Both methods give the same result: \$50,000

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Approach 3: Expenditure Approach

Concept: Sum spending by ultimate users of final goods and services

Final Good	Purchased By	Amount
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Fresh oranges		
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Orange juice		
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Total GDP		
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Think about:

- Who are the ultimate users of oranges?

Approach 3: Expenditure Approach

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Approach 3: Expenditure Approach

Concept: Sum spending by ultimate users of final goods and services

Final Good	Purchased By	Amount
Fresh oranges	Households	\$10,000
Orange juice	Households	\$40,000
Total GDP		\$50,000

Think about:

- Who are the ultimate users of oranges?
- Who are the ultimate users of orange juice?
- Is JuiceInc an ultimate user of oranges? Why or why not?
- What amounts did households spend on final goods?

Key Point: JuiceInc is **NOT** an ultimate user because it **trans-forms** oranges into juice for final consumption.

Rule: Only count spending on **final goods**, not intermediate goods.

GDP Identity: All Methods Give Same Result!

Approach	Result
----------	--------

Product Approach (Value Added)	
--------------------------------	--

Income Approach A (Before-tax)	
--------------------------------	--

Income Approach B (After-tax + taxes)	
---------------------------------------	--

Expenditure Approach	
----------------------	--

GDP Identity: All Methods Give Same Result!

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Fundamental Identity:

$$\text{GDP} = \text{Value Added} = \text{Income} = \text{Expenditure}$$

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Three different perspectives, same economic reality!

Key Concepts Review

Double Counting Problem:

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Three Perspectives:

- Product: What is produced (value added)
- Income: What is earned from production
- Expenditure: What is spent on final goods

Roadmap of Lecture

National Income Accounting

Gross Domestic Product

- * Juice Business Example: Calculation

Saving and Wealth

Real GDP, Price Indexes, and Inflation

- * Inflation
- * Application

Interest Rates

Problems

Wealth

- **Household wealth** = a household's assets minus its liabilities
- **National wealth** = sum of all households', firms', and governments' wealth within the nation
- Savings by individuals, businesses, and government determine wealth

Measures of aggregate saving

- **Saving** = current income – current spending
- **Saving rate** = saving / current income
- **Private saving** = private disposable income – consumption

$$S_{pvt} = (Y + NFP + TR + INT - T) - C \quad (2.6)$$

Measures of aggregate saving

- **Government saving** = net government income – government purchases of goods and services

$$S_{govt} = (T - TR - INT) - G \quad (2.7)$$

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- Government receipts = tax revenue (T)
- Government outlays = government purchases of goods and services (G) + transfers (TR) + interest payments on government debt (INT)

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- **Government budget deficit** = $-S_{govt}$

Measures of aggregate saving

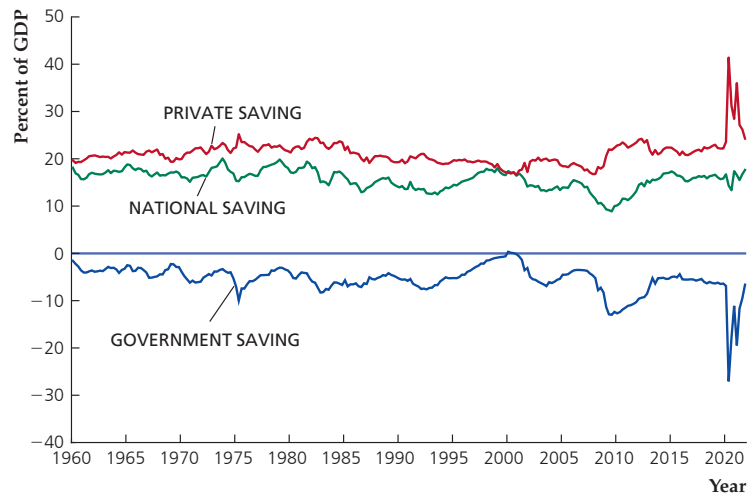
Simplification: count government investment as government purchases, not investment

■ **National saving** = private saving + government saving

$$S = S_{pvt} + S_{govt} \quad (2.8) \quad (2)$$

$$= [Y + NFP + TR + INT - C - T] + [T - TR - INT - G] \quad (3)$$

$$= Y + NFP - C - G = GNP - C - G \quad (4)$$

Figure 2.1: U.S. Saving Measures

The uses of private saving

$$S = I + (NX + NFP) \quad (2.9) \quad (5)$$

$$S = I + CA \quad (2.10) \quad (6)$$

■ Derived from $S = Y + NFP - C - G$ and $Y = C + I + G + NX$

■ $CA = NX + NFP$ = current account balance

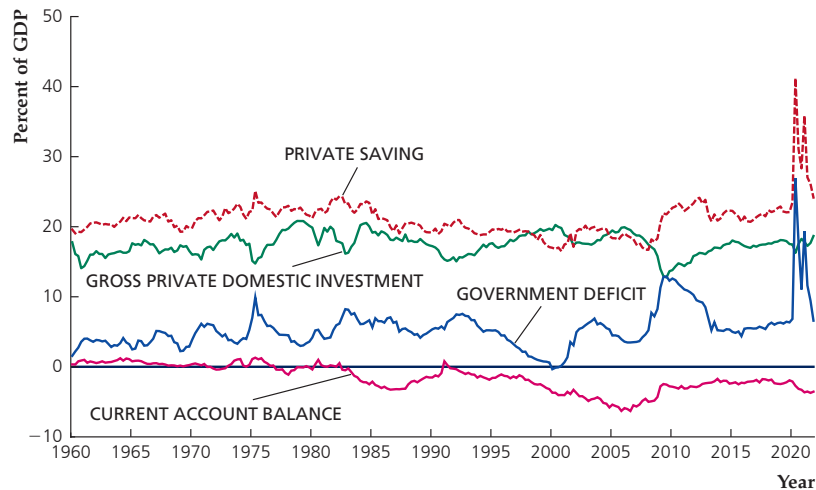
■ $S_{pvt} = I + (-S_{govt}) + CA$ (2.11)¹

¹(using $S = S_{pvt} + S_{govt}$)

The uses-of-saving identity

- investment (I)
- government budget deficit ($-S_{govt}$)
- current account balance (CA)

Components of the U.S. Uses-Of-Saving Identity



- The uses-of-saving identity (Fig 2.2)
 - Private saving, gross private domestic investment, government budget deficit, and current account balance, each as a percentage of GDP
 - Since early 1990s, current account balance has been negative
 - Foreigners buying U.S. assets
 - In some crisis periods, we observe large increases in the government budget deficit accompanied by sharp declines in gross private domestic investment and increases in private saving
 - In periods of strong growth, we observe a decline in private saving and the government budget deficit, along with a rise in investment

Relating Saving and Wealth

■ Stocks and flows

- **Flow variables:** measured per unit of time (GDP, income, saving, investment)
- **Stock variables:** measured at a point in time (quantity of money, value of houses, capital stock)
- Flow variables often equal rates of change of stock variables

- **Wealth and saving as stock and flow** (wealth is a stock, saving is a flow)

Relating Saving and Wealth

- **National wealth:** domestic physical assets + net foreign assets
- Country's domestic physical assets (capital goods and land)
- Country's net foreign assets = foreign assets (foreign stocks, bonds, and capital goods owned by domestic residents) minus foreign liabilities (domestic stocks, bonds, and capital goods owned by foreigners)
- Wealth matters because the economic well-being of a country depends on it

Relating Saving and Wealth (10 of 10)

National wealth: domestic physical assets + net foreign assets

■ Changes in national wealth

- Change in value of existing assets and liabilities (change in price of financial assets, or depreciation of capital goods)
- National saving ($S = I + CA$) raises wealth

■ Comparison of U.S. saving and investment with other countries

- The United States is a low-saving country; Japan is a high-saving country
- U.S. investment exceeds U.S. saving, so we have a negative current-account balance

SUMMARY 1

Measures of Aggregate Saving

Saving measure	Definition and formula
Private saving	Private disposable income less consumption $S_{\text{pvt}} = (Y + NFP + TR + INT - T) - C$
Government saving	Net government income - government purchases $S_{\text{govt}} = (T - TR - INT) - G$
National saving	Private saving plus government saving; also GNP $(Y + NFP)$ less consumption and government purchases $S = S_{\text{pvt}} + S_{\text{govt}}$ $= Y + NFP - C - G$

Roadmap of Lecture

National Income Accounting

Gross Domestic Product

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Interest Rates

Problems

Real GDP, Price Indexes, and Inflation (1 of 12)

- **Nominal variables** are those in dollar terms
- **Problem:** Do changes in nominal values reflect changes in prices or quantities?
- **Real variables:** adjust for price changes; reflect only quantity changes

Real GDP, Price Indexes, and Inflation (2 of 12)

- Example of computers and bicycles
- **Nominal GDP** is the dollar value of an economy's final output measured at current market prices
- **Real GDP** is an estimate of the value of an economy's final output, adjusting for changes in the overall price level

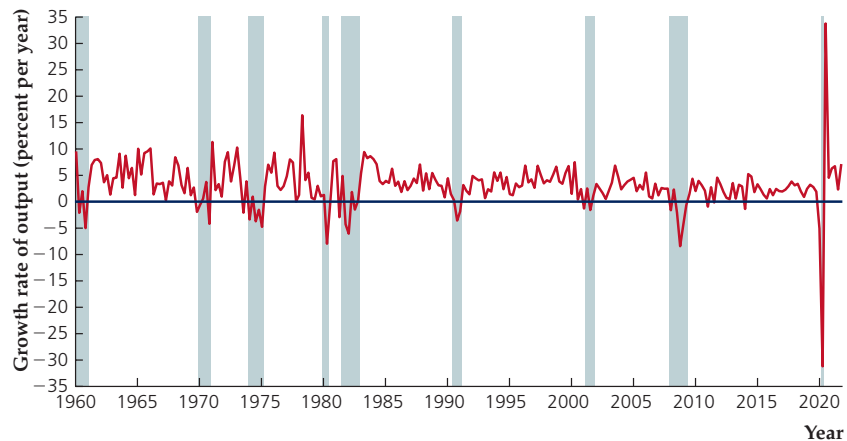
Real GDP, Price Indexes, and Inflation (3 of 12)

- Quarterly annualized growth rates of GDP are calculated using the formula

$$\frac{\Delta Y}{Y} = \left(\frac{Y(t)}{Y(t-1)} \right)^4 - 1 \quad (2.12)$$

- U.S. data are also generally seasonally adjusted
- Text Figure 2.3 shows quarterly annualized U.S. GDP growth rates

The Annualised Quarterly Growth Rate



The annualized quarterly growth rate of U.S. real GDP, 1960Q1–2021Q4 The quarterly growth rate of U.S. real GDP is annualized by convention to make comparisons between quarterly and annual data easier. The annualized quarterly growth rate is highly volatile and is frequently negative during recessions (shown by green shaded bars).

Table 2.3: Production and Price Data

	Year 1	Year 2	Percent change from year 1 to year 2
Product (quantity)			
Computers	5	10	+100%
Bicycles	200	250	+25%
Price			
Computers	\$1200/computer	\$600/computer	−50%
Bicycles	\$200/bicycle	\$240/bicycle	+20%
Value			
Computers	\$6000	\$6000	0
Bicycles	\$40,000	\$60,000	+50%
Total	\$46,000	\$66,000	+43.5%

Calculation of Real Output with Alt. Base Years

Calculation of real output with base year = Year 1					
	Current quantities		Base-year prices		
Year 1					
Computers	5	×	\$1200	=	\$6000
Bicycles	200	×	\$200	=	\$40,000
				Total =	\$46,000
Year 2					
Computers	10	×	\$1200	=	\$12,000
Bicycles	250	×	\$200	=	\$50,000
				Total =	\$62,000
Percentage growth of real GDP = $(\$62,000 - \$46,000)/\$46,000 = 34.8\%$					
Calculation of real output with base year = Year 2					
	Current quantities		Base-year prices		
Year 1					
Computers	5	×	\$600	=	\$3000
Bicycles	200	×	\$240	=	\$48,000
				Total =	\$51,000
Year 2					
Computers	10	×	\$600	=	\$6000
Bicycles	250	×	\$240	=	\$60,000
				Total =	\$66,000
Percentage growth of real GDP = $(\$66,000 - \$51,000)/\$51,000 = 29.4\%$					

Price Indexes

A price index measures the average level of prices for some specified set of goods and services, relative to the prices in a specified base year

- Calculate inflation rate:

$$\pi_{t+1} = (P_{t+1} - P_t)/P_t = \Delta P_{t+1}/P_t$$

- Two measures of π_{t+1} : GDP deflator and CPI.

GDP deflator

■ **GDP deflator** = $100 \times \text{nominal GDP} / \text{real GDP}$

■ Note that base year $P = 100$

■ Inflation = growth rate of GDP deflator

1. GDP deflator is an index, without economic interpretation
2. The index structure emerges from the mathematical nature of the calculation itself, not from any economic theory about what price levels should look like or how they should behave.

Price Indexes

Consumer Price Index (CPI)

- Monthly index of consumer prices; index averages 100 in reference base period (1982 to 1984)
- Based on basket of goods in expenditure base period (updated periodically)

THE CPI BASKET: EXPENDITURE SHARES

for 2012 Base Year <https://public.flourish.studio/visualisation/13807161/>

Price Indexes

The computer revolution and chain-weighted GDP

- Choice of expenditure base period matters for GDP when prices and quantities of a good, such as computers, are changing rapidly

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Price Indexes

The computer revolution and chain-weighted GDP

- Choice of expenditure base period matters for GDP when prices and quantities of a good, such as computers, are changing rapidly
- BEA compromised by developing chain-weighted GDP
- Now, however, components of real GDP don't add up to real GDP, but discrepancy is usually small

Quantity Index Formulas

Laspeyres Quantity Index (Base Year Prices) Uses **base year prices** as weights:

$$\text{Laspeyres Quantity Index} = \frac{P_0 \cdot Q_1}{P_0 \cdot Q_0}$$

Paasche Quantity Index (Current Year Prices) Uses **current year prices** as weights:

$$\text{Paasche Quantity Index} = \frac{P_1 \cdot Q_1}{P_1 \cdot Q_0}$$

Laspeyres Quantity Index (Base Year Prices) Uses **base year prices** to value output in different years:

$$L_t = \sum p_0 \cdot q_t$$

where p_0 = base year prices

q_t = quantities in year t

Paasche Quantity Index (Current Year Prices) Uses **current**

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q_t = quantities in year t

Chain-Weighted Index Formula

The Problem

- Laspeyres index may **overstate** growth (uses old, high prices)

Chain-Weighted Solution Take the **geometric mean** of both indexes:

$$\begin{aligned}C_t &= \sqrt{L_t \times P_t} \\ &= (L_t \times P_t)^{1/2}\end{aligned}$$

This provides a **compromise** between the two methods.

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Chain-Weighted Index Formula

The Problem

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- Paasche index may **understate** growth (uses new, low prices)
- Problem is severe when prices change rapidly (like computers in 1990s)

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Example: Computer-Bicycle Economy

Given Data from Table 2.4 [▶ Back](#)

- $L_1 = \$46,000$ (Year 1 output at Year 1 prices)

Chain-Weighted Calculations

$$C_1 = \sqrt{L_1 \times P_1} = \sqrt{46,000 \times 51,000} = \$48,400$$

$$C_2 = \sqrt{L_2 \times P_2} = \sqrt{62,000 \times 66,000} = \$63,970$$

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Growth Rate Comparison

Chain-Weighted Growth Rate

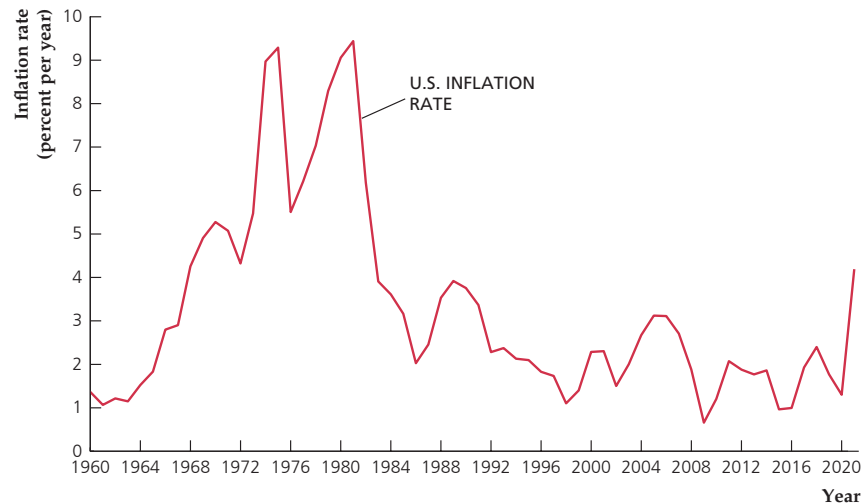
$$\begin{aligned}\text{Growth Rate} &= \frac{C_2 - C_1}{C_1} = \frac{63,970 - 48,400}{48,400} \\ &= \frac{15,570}{48,400} = 32.06\%\end{aligned}$$

Comparison with Traditional Methods

Method	Growth Rate
Laspeyres	$\frac{62,000 - 46,000}{46,000} = 34.8\%$
Paasche	$\frac{66,000 - 51,000}{51,000} = 29.4\%$
Simple Average	$\frac{34.8\% + 29.4\%}{2} = 32.1\%$
Chain-Weighted	32.06%

Key Insight: Chain-weighting gives result very close to simple average!

The inflation rate in the United States, 1960–2021



The Inflation Rate in the United States, 1960–2021 Here, inflation is measured as the annual percentage change in the GDP deflator. Inflation rose during the 1960s and 1970s, fell from the early 1980s through the 1990s, and has neither trended upward nor downward since then.

Figure: 2.4

Does CPI inflation overstate $\uparrow$ in the cost of living?

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- This problem is known as **substitution bias**
- If inflation is overstated, then real incomes are higher than we thought and we have overindexed payments like Social Security

Latest research suggests bias is still 1% per year or highe

CPI vesus GDP Deflator

- discrepancy 1: goods produced in the economy (GDP) are different from goods purchased by consumers (CPI)
 - some goods in GDP are sold to firms (I), the government (G), or foreigners (NX), not to consumers
 - some goods in CPI are imported from abroad, not produced domestically

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CPI versus GDP Deflator

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 - because the composition of GDP varies over time
- **YET, CPI AND GDP DEFLATOR MOVE TOGETHER!!**

Application: The Fed's preferred inflation measures

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- Differences between the PCE price index and the CPI include formulas used in their calculation, coverage of different items, and weights given to different items

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- But the Fed forecasts both the core and overall PCE price index because the Fed needs to keep its eye on both underlying trends but also the actual inflation rate faced by households
- The inflation rate in the overall PCE price index tends to revert to the core measure after a period when the two measures deviate (Figure 2.5)

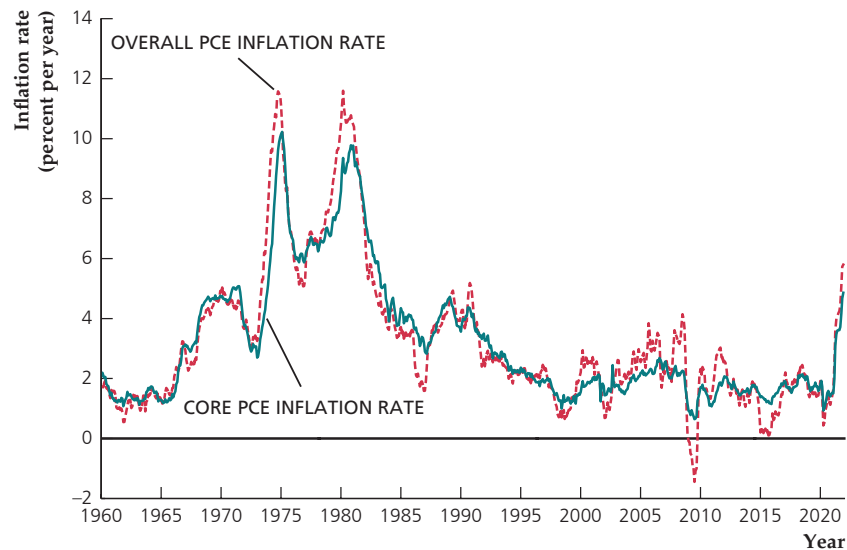
Figure 2.5: Overall PCE Inflation Rate and Core PCE

Figure: January 1960 to December 2021

Why Do We Care About Inflation: Classical Reasons

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- **Long-term contracts:** Some prices and wages are fixed by long-term contracts (for instance, some loans or pension payments)
- **Tax system distortion:** Inflation distorts the tax system because income brackets are fixed
- **Uncertainty:** Inflation creates uncertainty in economic planning

Why Do We Care About Inflation: Psychological Reasons

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- **Real-world example:** `https://www.nytimes.com/2014/03/14/world/americas/rocketing-prices-confound-an-argentina-racked-by-`
`html`

Roadmap of Lecture

National Income Accounting

Gross Domestic Product

- * Juice Business Example: Calculation

Saving and Wealth

Real GDP, Price Indexes, and Inflation

- * Inflation
- * Application

Interest Rates

Problems

Interest Rates (1 of 3)

■ Real vs nominal interest rates

- **Interest rate:** a rate of return promised by a borrower to a lender
- **Real interest rate:** rate at which the real value of an asset increases over time
- **Nominal interest rate:** rate at which the nominal value of an asset increases over time

Interest Rates (2 of 3)

- **Real vs nominal interest rates**

$$\text{Real interest rate} = i - \pi \quad (2.13)$$

- Text Figure 2.6 plots nominal and real interest rates for the United States since 1960

Nominal and Real Interest Rates in the US: 1960–2021

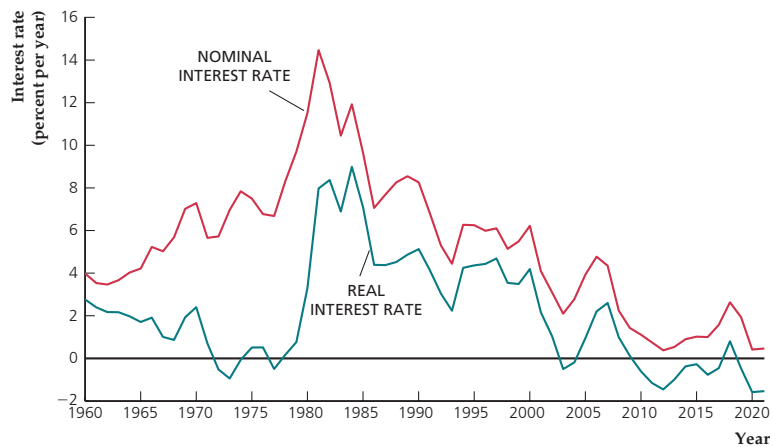


Figure: 2.6 ABC

Interest Rates (3 of 3)

■ The expected real interest rate

$$r = i - \pi^e \quad (2.14)$$

If $\pi = \pi^e$, real interest rate = expected real interest rate

Summary

■ Key concepts learned:

- Three approaches to national income accounting (all equivalent)
- GDP measurement: $Y = C + I + G + NX$
- Saving-investment identity: $S = I + CA$
- Real vs nominal variables and price measurement
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- They must hold by definition, providing a solid base for economic modeling

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Interest Rates

Problems

1. In the country of Kwaki, people produce canoes, fish for salmon, and grow corn. In one year, they produced 5000 canoes using only labor and natural materials but sold only 4000 as the economy had entered a recession. The cost of producing each canoe was \$1000, but the ones that sold were priced at \$1250. They fished \$30 million worth of salmon. They used \$3 million worth of the salmon as fertilizer for corn. They grew and consumed \$55 million worth of corn. What was Kwaki's GDP for the year?

- **Canoes = \$6,000,000**
 - Sold canoes: $4,000 \times \$1,250 = \$5,000,000$
 - Unsold inventory: $1,000 \times \$1,000 = \$1,000,000$
 - *(Unsold inventory counts as investment at production cost)*
- **Salmon = \$27,000,000**
 - Only the \$27,000,000 not used as fertilizer counts
 - *The \$3,000,000 used as fertilizer is an intermediate good*
- **Corn = \$55,000,000**
 - Full value of final corn production
 - *Already incorporates the value of salmon fertilizer input*

Total GDP = \$88,000,000

Problem: 3

ABC Computer Company has a \$1.4 billion (Indian rupees) factory in Bangalore, India. During the current year, ABC builds \$140 million worth of computer components. ABC's costs for labor are \$70 million; interest on debt, \$7 million; and taxes, \$14 million. ABC sells all its output to XYZ Supercomputer. Using ABC's components, XYZ builds four supercomputers at a cost of \$56 million each (components worth \$35 million, labor costs of \$14 million, and \$7 million in taxes per computer). XYZ has a \$2.1 billion factory. XYZ sells three of the supercomputers for \$70 million each. At year's end, it had not sold the fourth. The unsold computer is carried on XYZ's books as a \$56 million increase in inventory.

1. Calculate the contributions to GDP of these transactions, showing that all three approaches give the same answer.
2. Repeat part (a) , but now assume that, in addition to its other costs, ABC paid \$35 million for imported computer chips.

ABC produces output valued at \$140 million and has total expenses of \$91 million (\$70 million for labor, \$7 million in interest, and \$14 million in taxes), so its profits are \$49 million. XYZ produces output valued at \$266 million (\$210 million for the three computers that were sold and \$56 million for the unsold computers in the inventory) and has expenses of \$224 million (\$140 million for components, \$56 million for labor, and \$28 million for taxes). So its profits are \$42 million.

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1. Calculate the contributions to GDP of these transactions, showing that all three approaches give the same answer.
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According to the product approach, the GDP contributions of these companies are \$266 million – the value of the final product of XYZ. ABC's production is of an intermediate good, used completely by XYZ, and so is not part of the GDP. According to the expenditure approach, the GDP contribution is also \$266 million, with \$210 million (of sold computers) adding to the capital stock (as investment spending), and \$56 million (the unsold computers) as inventory investment.

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Table: Income Approach to GDP Calculation

Component	ABC	XYZ	TOTAL
Labor	\$70 million	\$56 million	\$126 million
Profit	\$49 million	\$42 million	\$91 million
Taxes	\$14 million	\$28 million	\$42 million
Interest	\$7 million	\$0 million	\$7 million
Total	\$140 million	\$126 million	\$266 million

Problem: 5

- Gross private domestic investment = 40
- Government purchases of goods and services = 30
- Gross national product (GNP) = 200
- Current account balance = -20
- Taxes = 60
- Government TR payments to the domestic private sector = 25
- Interest payments from the government to the domestic private sector = 15. (Assume all INT payments (Govt) $\Rightarrow$ domestic Hhs.)
- Factor income received from rest of world = 7
- Factor payments made to rest of world = 9

Find the following, assuming that government investment is zero:

- | | |
|--------------------|-------------------|
| 1. C | 5. S_{pvt} |
| 2. NX | 6. S_{Govt} |
| 3. GDP | 7. $S_{National}$ |
| 4. NFP from abroad | |

$$NFP = 7 - 9 = -2 \quad (7)$$

$$GDP = GNP - NFP = 200 - (-2) = 202 = Y \quad (8)$$

$$NX = CA - NFP = -20 - (-2) = -18 \quad (9)$$

$$C = Y - (I + G + NX) = 202 - (40 + 30 + (-18)) = 150 \quad (10)$$

$$S_{pvt} = (Y + NFP - T + TR + INT) - C \quad (11)$$

$$= (202 + (-2) - 60 + 25 + 15) - 150 = 30 \quad (12)$$

$$S_{govt} = (T - TR - INT) - G \quad (13)$$

$$= (60 - 25 - 15) - 30 = -10 \quad (14)$$

$$S = S_{pvt} + S_{govt} = 30 + (-10) = 20 \quad (15)$$

Problem: 8

Loretta agrees to lend Ted \$500,000 to buy computers for his consulting firm. They agree to a nominal interest rate of 8%. Both expect the inflation rate to be 2%.

1. Calculate the expected real interest rate.
2. If inflation turns out to be 3% over the life of the loan, what is the real interest rate? Who gains from unexpectedly high inflation, Loretta or Ted?
3. If inflation turns out to be 1% over the life of the loan, what is the real interest rate? Who gains from unexpectedly low inflation, Loretta or Ted?

(a) Expected Real Interest Rate

The expected real interest rate is calculated as:

$$\text{Expected real interest rate} = \text{Nominal rate} - \text{Expected inflation rate} \quad (16)$$

$$= 8\% - 2\% \quad (17)$$

(b) Scenario: High Inflation (3%)

The actual real interest rate is:

$$\text{Real interest rate} = 8\% - 3\% = 5\% \quad (18)$$

Ted gains from the unexpectedly high inflation because he repays the loan with dollars that aren't worth as much as he expected.

(c) Scenario: Low Inflation (1%)

The actual real interest rate is:

$$\text{Real interest rate} = 8\% - 1\% = 7\% \quad (19)$$

Loretta gains from the unexpectedly low inflation because she gets repaid with dollars that are worth more than she expected.

Thank You for Your Attention!

*“I know you think you understand what you thought I said,
but I’m not sure you realize that what you heard is not what I
meant.”*

– Alan Greenspan